International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Student Practical Laboratory Manual

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Course code: CIP-B101

Lab title: PC System Fundamentals, Disk Sectors and Partition Analysis

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Instructor name: ABDUL

Introduction

This lab introduces the basic computer system knowledge required for beginner digital forensic work. The student will identify system hardware and software information, map where digital evidence can exist, calculate disk and partition sizes from sector information, inspect partition layout using safe read-only commands, explain the boot process, and identify file systems used by common operating systems.

- Tools used: list operating system, commands, and any software used.

Tools used

Operating systems	Commands	Software
Windows		winhex

6. Lab Scenario

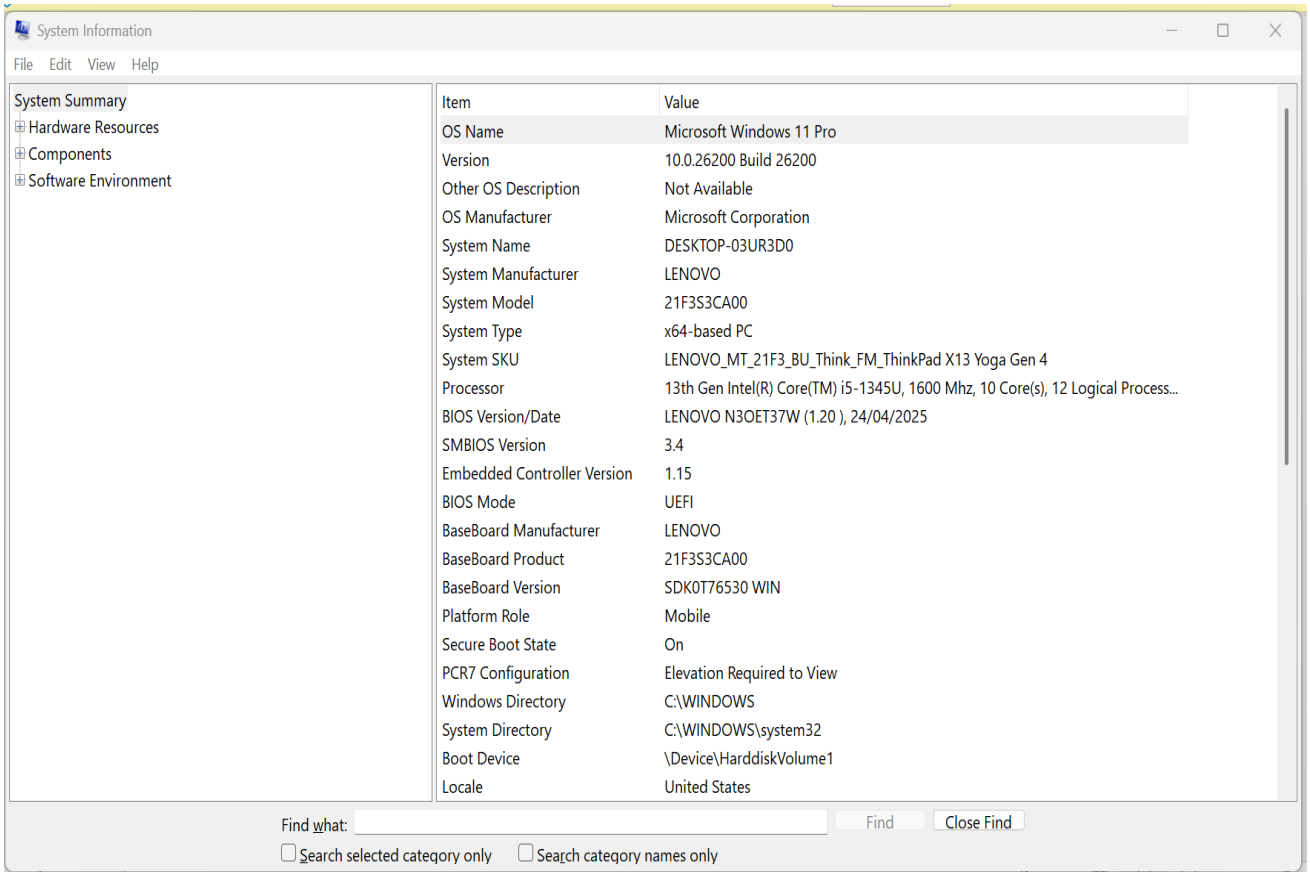
You are a junior digital forensic trainee asked to prepare an initial system profile for a computer that may later be used in a forensic investigation. Your task is to document the system safely, identify possible evidence layers, inspect disk and partition details, calculate partition sizes, explain the boot process, and identify file systems without altering the computer.

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

A1. Capture basic system specification

- On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.
  Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.



- Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```

```
C:\Users\USER>systeminfo

Host Name:                 DESKTOP-03UR3D0
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          USER
Registered Organization:
Product ID:                00355-61319-80397-AAOEM
Original Install Date:     26/06/2026, 17:34:16
System Boot Time:          23/10/2026, 14:13:45
System Manufacturer:       LENOVO
System Model:              21F3S3CA00
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 186 Stepping 3 GenuineIntel ~1066 Mhz
BIOS Version:              LENOVO N3OET37W (1.20 ), 24/04/2025
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+03:00) Nairobi
Total Physical Memory:     16,041 MB
Available Physical Memory: 3,140 MB
Virtual Memory: Max Size:  28,940 MB
```

A2. Capture storage information

- Open Command Prompt or PowerShell.
- Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.
- If a command is not supported on your system, use the alternative command.

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
Get-Partition
Get-Volume
```

```
PS C:\WINDOWS\System32> Get-disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                            Style
------ ------------ -------------                  ------------         -----------------      ---------- ----------
0      SKHynix_HF... 0000_0000_0000_0000_ACE4_2E00... Healthy            Online                  476.94 GB GPT
```

```
PS C:\WINDOWS\System32> Get-partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs512ge#5&4085307&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                 248.73 GB Basic
4                            267198136320                                1.02 GB Recovery
5                D           268292849664                              100 GB Basic
6                E           375667032064                              100 GB Basic
```

```
PS C:\WINDOWS\System32> Get-volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    84.19 MB   1.02 GB
C                        NTFS           Fixed     Healthy      OK                   107.04 GB 248.73 GB
D           DHI          NTFS           Fixed     Healthy      OK                    98.68 GB    100 GB
E           KITYA        NTFS           Fixed     Healthy      OK                    94.17 GB    100 GB
```

Evidence layer mapping

Table below shows system layers/sources and the type of evidence that can be generated from each one.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behavior

Evidence may exist in more than one system layer because a single user activity can generate several digital traces across different components of a computer system. For example, when a user downloads and opens a document:

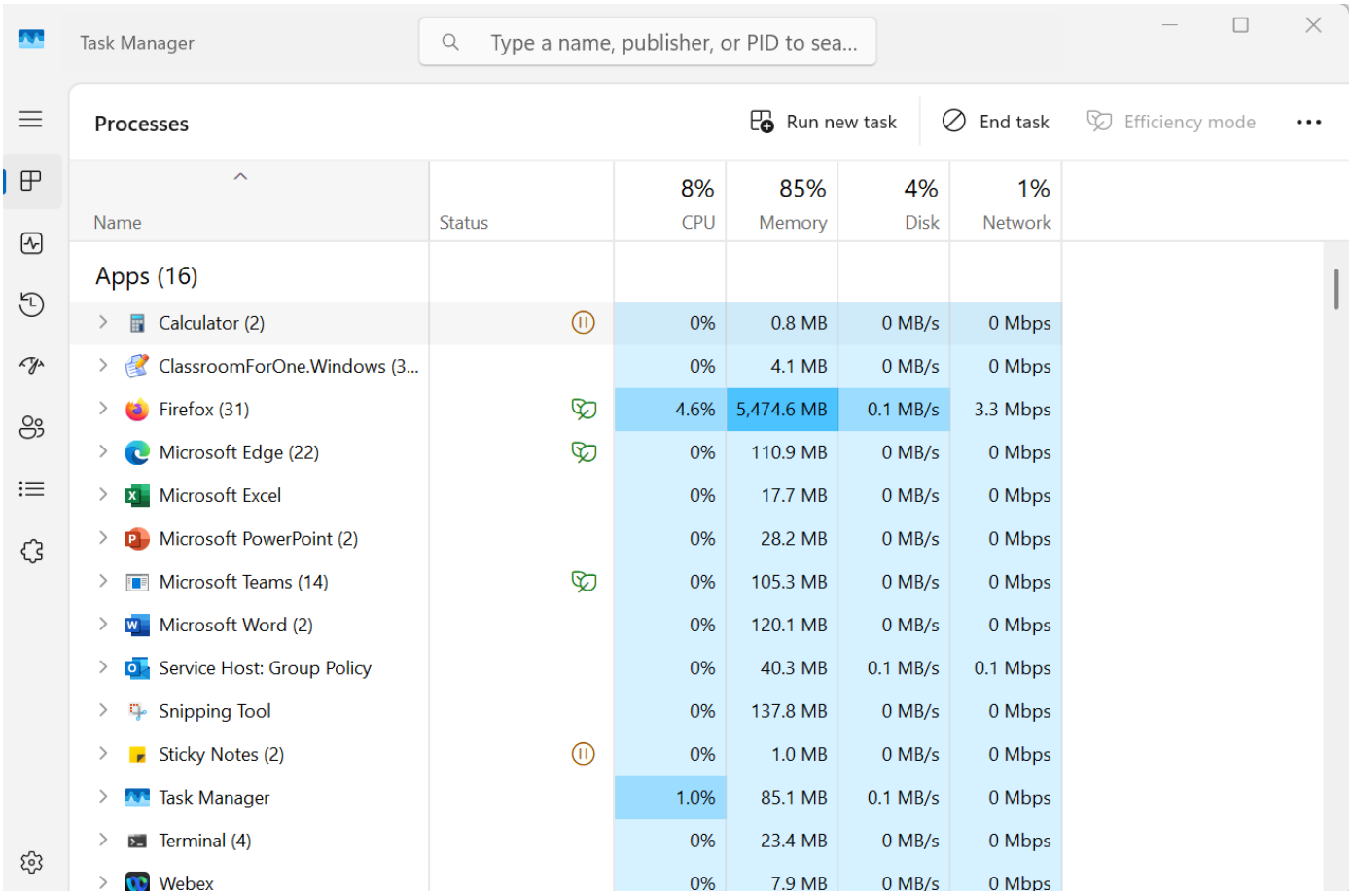
- The user layer may show the browsing activity.
- The application layer may retain the download history.
- The operating system may record related file activity.
- The disk layer stores the downloaded document.
- The memory layer may contain information about the running application and open document.

3. Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation.

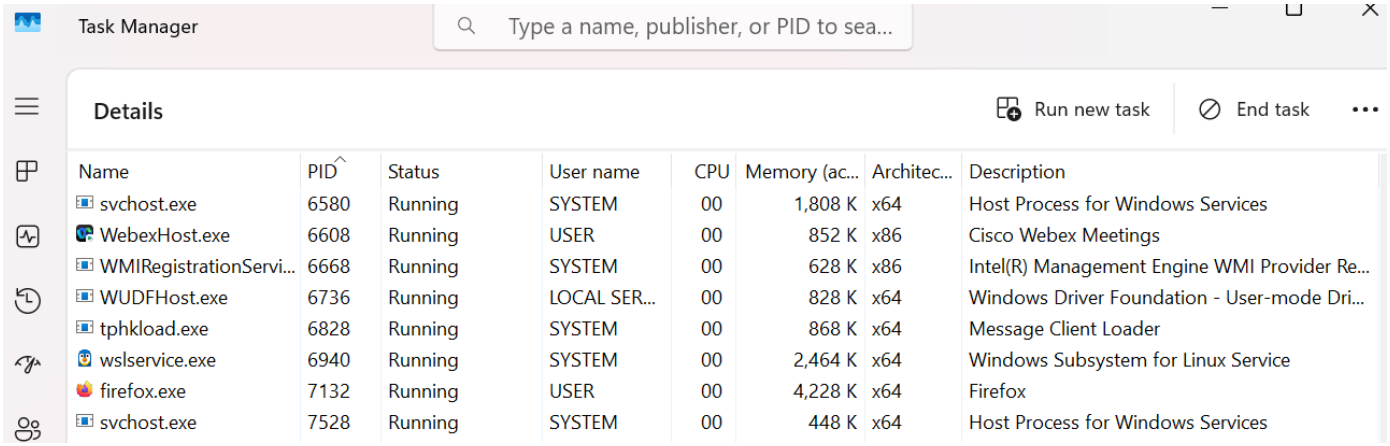
3.1 Running processes

The figure below shows the running processes on my computer.



3.1.2 Running processes

The figure below shows sample processes, their process names, PID values, and likely purpose/ description.



Name	PID	Status	User name	CPU	Memory (ac...	Architec...	Description
svchost.exe	6580	Running	SYSTEM	00	1,808 K	x64	Host Process for Windows Services
WebexHost.exe	6608	Running	USER	00	852 K	x86	Cisco Webex Meetings
WMIRegistrationServi...	6668	Running	SYSTEM	00	628 K	x86	Intel(R) Management Engine WMI Provider Re...
WUDFHost.exe	6736	Running	LOCAL SER...	00	828 K	x64	Windows Driver Foundation - User-mode Dri...
tphkload.exe	6828	Running	SYSTEM	00	868 K	x64	Message Client Loader
wslservice.exe	6940	Running	SYSTEM	00	2,464 K	x64	Windows Subsystem for Linux Service
firefox.exe	7132	Running	USER	00	4,228 K	x64	Firefox
svchost.exe	7528	Running	SYSTEM	00	448 K	x64	Host Process for Windows Services

3.2 Program, Process and Thread

A program is a set of instructions written to perform a specific task. It is usually stored on a computer's disk and does not necessarily need to be running. E.g Google Chrome installed on a computer (`chrome.exe`).

A process is a program that is currently executing. When a user opens Google Chrome, the operating system creates a process and allocates resources such as memory and CPU time to it. Each process is assigned a unique Process ID (PID). E.g When Chrome is running, Windows may show `chrome.exe` with a specific PID in Task Manager or the `tasklist` command output.

A thread is the smallest unit of execution within a process. A process can have multiple threads working on different tasks simultaneously while sharing the process's resources. E.g Google Chrome uses multiple threads to handle activities such as loading web pages, processing user input and displaying content.

3.3 Use of process information during malware analysis or live forensic triage.

Process information helps investigators understand what is currently running on a computer and identify potentially suspicious activities. By examining process names, Process Identification Numbers (PIDs), resource usage and running applications, investigators can identify unfamiliar or suspicious processes.

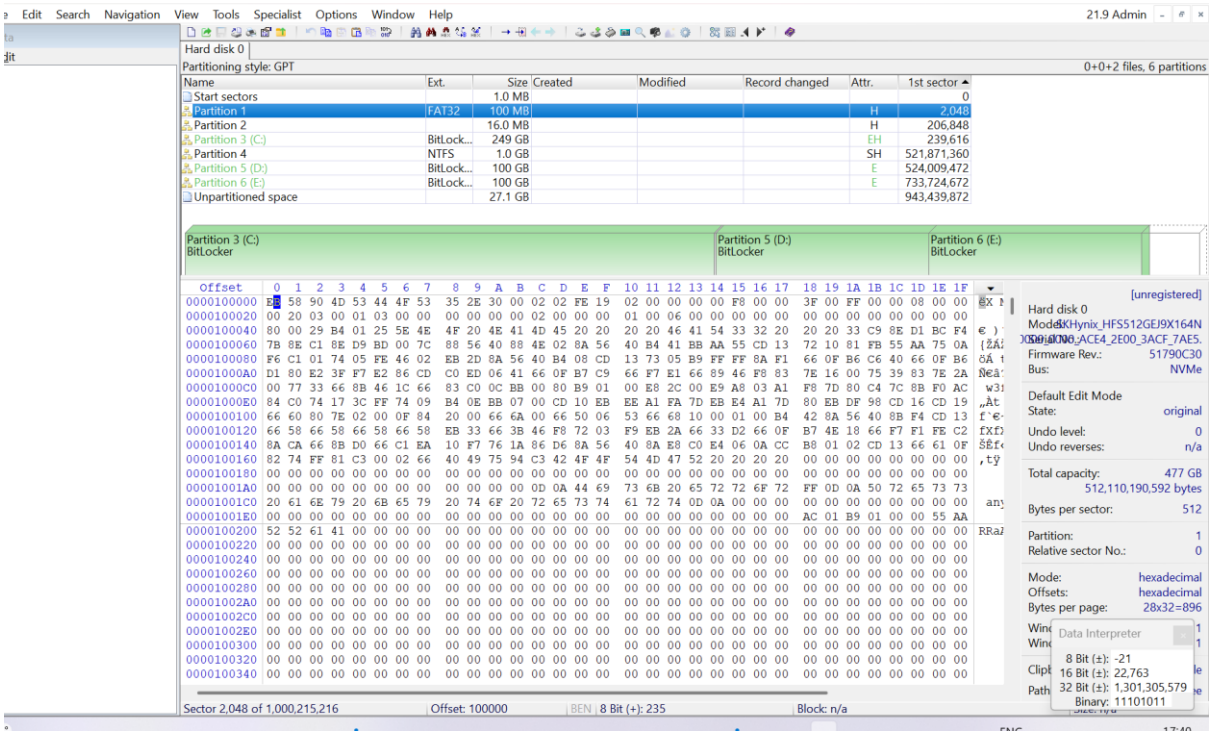
It also helps investigators to:

1. Identify potentially malicious programs running in the background.
2. Detect unusual CPU and memory usage.
3. Identify suspicious network connections associated with running processes.
4. Understand which applications or services are active.
5. Collect evidence of system activity during live forensic investigations.

4. Disk Partition Inspection

C2. Calculate your own disk or partition size

The figure below represents partitions for my hard disk and for purposes of this study, we shall use partition 1 and 4



- For each partition, record the start sector, number of sectors, sector size, and file system where available.

Partition	Start sector	Start sector for next partition	No of sectors	File system
1	2048	206848	206848-2048 = 204800	FAT 32
4	521871360	524009472	524009472-521871360 =2138112	NTFS

- Calculate the size of each partition in GB using the formula above.
- Your calculation must be shown step-by-step. Screenshots alone are not enough.

Sector size = 512 bytes
Partition size = No of sectors * Bytes per sector/ 1024/1024/1024

Partition 1 = 204800*512/1024/1024/1024

=0.09765625

Approximately: 0.098GB

Partition 4 = 2138112*512/1024/1024/1024

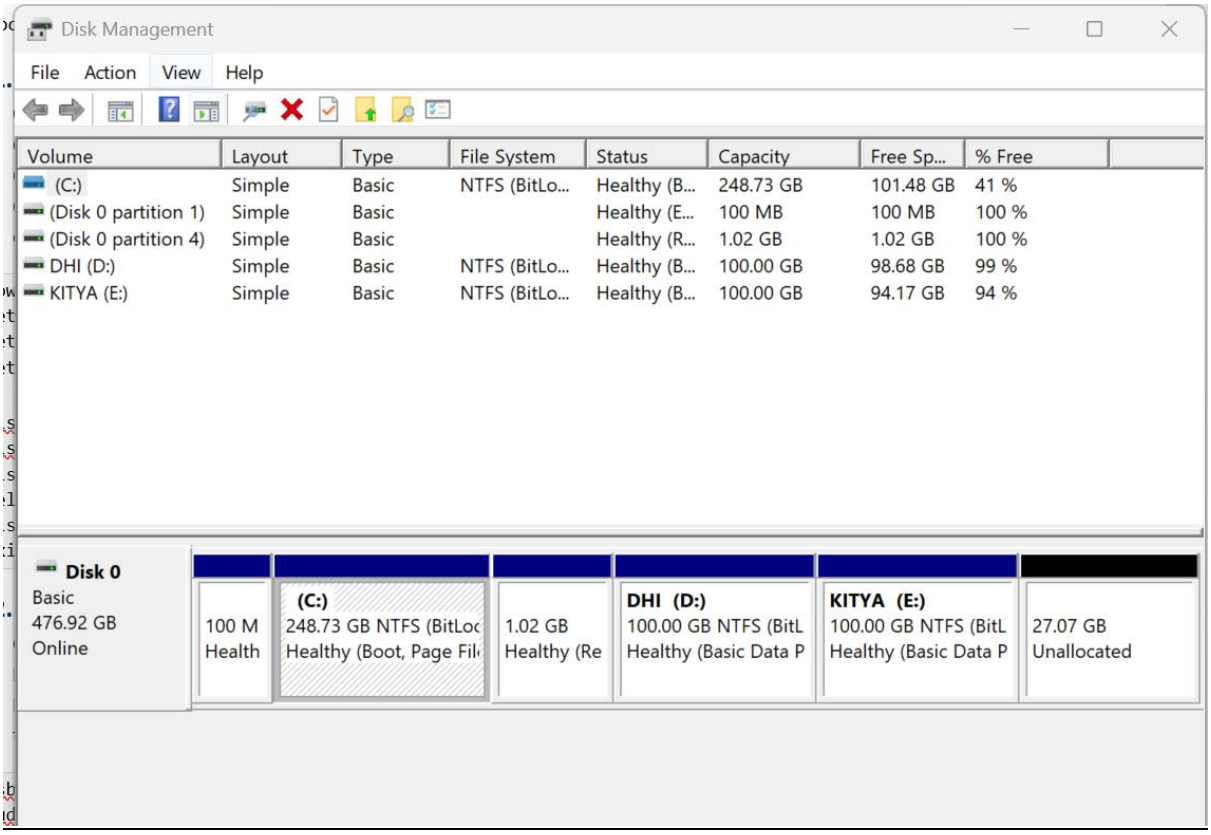
= 1.01953

Approximately 1.02GB

4.1 Safe Partition Table Inspection

D1. Windows method - Disk Management, PowerShell and DiskPart

- Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.
- Capture a screenshot showing your disk layout and visible partitions.



- Open PowerShell as normal user or administrator if required and run the safe read-only commands below.
- Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

```
PowerShell:
Get-Disk
Get-Partition
Get-Volume

DiskPart safe listing only:
diskpart
list disk
select disk 0
list partition
exit
```

```
PS C:\WINDOWS\System32> Get-Disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                           Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      SKHynix_HF... 0000_0000_0000_0000_ACE4_2E00... Healthy             Online                  476.94 GB GPT


PS C:\WINDOWS\System32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs512ge#5&4085307&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                 248.73 GB Basic
4                            267198136320                                1.02 GB Recovery
5                D           268292849664                              100 GB Basic
6                E           375667032064                              100 GB Basic
```

```
PS C:\WINDOWS\System32> Get-volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    84.19 MB   1.02 GB
C                        NTFS           Fixed     Healthy      OK                   101.48 GB 248.73 GB
D           DHI          NTFS           Fixed     Healthy      OK                    98.68 GB    100 GB
E           KITYA        NTFS           Fixed     Healthy      OK                    94.17 GB    100 GB

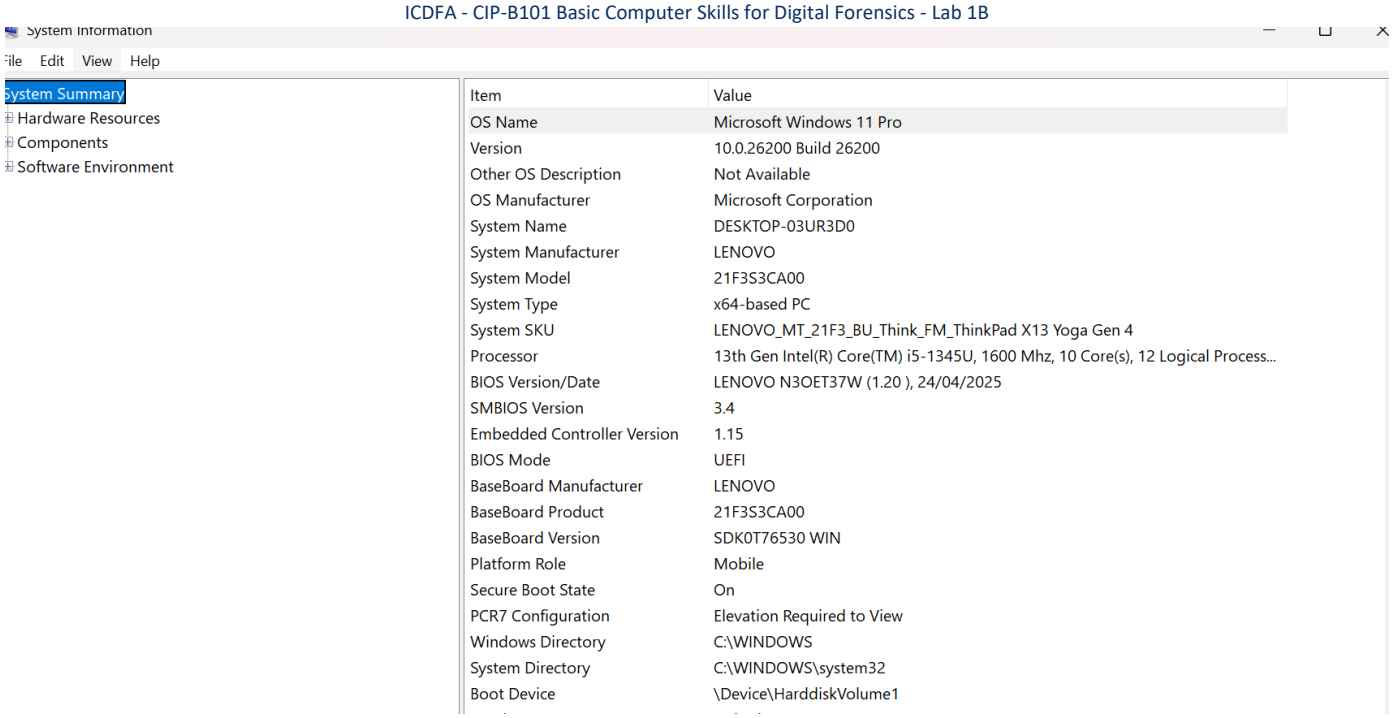

PS C:\WINDOWS\System32>
```

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

Power on (Computer receives power) -------> POST (Hardware checks are performed) ----------> UEFI Firmware (Firmware and CMOS settings applied) -------> EFI system partition (UEFI locates boot entry and EFI boot files) -------> Windows boot manager (Initiates windows) ----------> Window kernel (Kernel and windows loaded) ------> windows starts and becomes ready for use.

- State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.
  It is UEFI as seen in the attachment.



Stage	What Happens	Forensic Relevance
Power-on and POST	Hardware is initialized and checked.	Hardware failure or tampering may affect evidence access.
BIOS/UEFI firmware	Boot device order and firmware settings are applied.	Boot mode can explain partition style and boot artifacts.
MBR/GPT or EFI boot	System locates bootable partition or EFI boot entry.	Boot records may contain evidence of boot configuration.
Boot loader/manager	OS selection and pre-kernel loading occurs.	May show multi-boot or hidden OS behavior.
Kernel and OS control	Operating system fully starts.	Logs and user activity begin to accumulate.

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s.	Modern partitioning system used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.

Part F - File System Identification and Interpretation

- Identify the file system used by at least two volumes or partitions on your system.
- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.

- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.

  Partition 1: FAT32
  Partition 4: NTFS

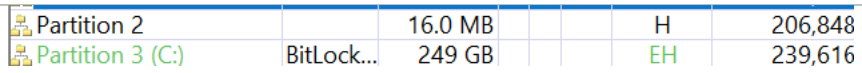
- In your report, explain why a file system is important in digital forensics.

  A file system is important in digital forensics because it determines how data is organized, stored, accessed and managed on a storage device. It helps forensic investigators locate files, examine file properties, identify timestamps, recover deleted files and understand how data was used.

Part G - Final Practical Assignment

Main assignment

Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

Task	Required Evidence	Student Notes / Answer
Identify partition 1		File format is FAT32 and start sector is 2048
Record partition 1 sector count	Start sector =2048, end sector = 206848 or sector count =204800	
Calculate partition 1 size	Formula: sector count *512/1024/1024/1024 Result =0.09765625 GB	
Identify partition 2		
Record partition 2 sector count	Start sector=206848, end sector =239616 sector count=32768	
Calculate partition 2 size	Formula : sector count *512/1024/1024/1024 result =0.015625 GB	

8. Student Report Template

- Cover page: student name, registration number, course code, lab title, date, and instructor name.
- Introduction: state the purpose of the lab in 5-8 lines.
- Tools used: list operating system, commands, and any software used.
- System specification evidence: include screenshots and explain important observations.
- Evidence layer mapping: include the required table.

- Process observation: include process/PID evidence and explanation.
- Disk and partition inspection: include partition screenshots and command output.
- Manual calculations: show formulas and results for at least two partitions.
- Boot process explanation: include your diagram or numbered boot sequence.
- File system identification: explain the file systems found.
- Conclusion: summarize what you learned and why these matters in digital forensics.
- Appendix: include extra screenshots if necessary.

9. Screenshot Checklist

No.	Screenshot Required	Included?
1	System Information or systeminfo output	Yes / No
2	Disk drive model and size output	Yes / No
3	Logical volume/file system output	Yes / No
4	Running process/PID output	Yes / No
5	Disk Management, Get-Partition, DiskPart, lsblk, fdisk, or parted output	Yes / No
6	Partition 1 calculation evidence	Yes / No
7	Partition 2 calculation evidence	Yes / No
8	Boot mode evidence: BIOS/UEFI	Yes / No
9	File system identification evidence	Yes / No
10	Final report summary/conclusion page	Yes / No

10. Marking Rubric - 100 Marks

Assessment Area	Marks	What Instructor Will Check
Preparation and safety compliance	10	Student followed read-only approach and did not perform destructive actions.
System specification and evidence layer mapping	15	Clear screenshots and explanation of system layers and evidence sources.
Program/process/thread explanation	10	Correct process evidence and explanation of forensic relevance.
Disk sector and size calculation	20	Accurate formulas, calculations, unit conversions, and explanation.
Partition table inspection	20	Reliable partition evidence from approved commands/tools; two partitions analyzed.
Boot process and MBR/GPT explanation	10	Correct boot sequence and comparison of MBR/GPT.
File system identification	10	Correct file system evidence and interpretation.

Assessment Area	Marks	What Instructor Will Check
Report quality and screenshots	5	Professional formatting, clear screenshots, and complete submission.

11. Common Mistakes to Avoid

- Submitting screenshots without manual calculation steps.
- Using only total disk size and failing to identify partitions 1 and 2.
- Confusing GB and GiB rounding differences without explanation.
- Running destructive disk commands such as clean, format, delete, mkfs, or dd to a physical disk.
- Failing to explain why file systems matter in digital forensics.
- Copying slide definitions without adding your own observations and screenshots.

12. Source Acknowledgement

This lab was prepared from the course slide material titled Computer System Introduction for CIP-B101 Basic Computer Skills for Digital Forensics. Students are expected to read the slide material first because it provides the walkthrough for the practical work.